SMC-CS79F-Module2-Lab2-NOTEBOOK-DerekTarvin

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Derek Tarvin SMC-CS79F-1755 Fall 2026 Module2-Lab2-DerekTarvin.ipynb GitHub portfolio repository URL: https://github.com/nostradomus00/aws-ai-portfolio

[1]:
```
import pandas as pd

df = pd.read_csv('m2_metric_scenarios.csv')

tp = df['true_positive']
fp = df['false_positive']
fn = df['false_negative']
tn = df['true_negative']

df['my_accuracy'] = (tp + tn) / (tp + fp + fn + tn)
df['my_precision'] = tp / (tp + fp)
df['my_recall'] = tp / (tp + fn)
df['my_f1'] = (2 * tp) / (2 * tp + fp + fn)

# compare my calculated metrics with the csv file's precomputed columns
display(df[[
    'scenario_id',
    'accuracy', 'my_accuracy',
    'precision', 'my_precision',
    'recall', 'my_recall',
    'f1', 'my_f1'
]].round(4))
```

```
  scenario_id  accuracy  my_accuracy  precision  my_precision  recall  \
0          S1    0.9900       0.9900     0.8333        0.8333  0.9524
1          S2    0.9918       0.9918     0.9266        0.9266  0.7947
2          S3    0.9930       0.9930     0.8757        0.8757  0.9226
3          S4    0.9536       0.9536     0.8000        0.8000  0.8455

   my_recall      f1   my_f1
0     0.9524  0.8889  0.8889
1     0.7947  0.8556  0.8556
2     0.9226  0.8986  0.8986
3     0.8455  0.8221  0.8221
```

[2]:
```
df['total_error_cost'] = (
    df['false_positive'] * df['dollar_cost_per_false_positive']
    + df['false_negative'] * df['dollar_cost_per_false_negative']
)

# show highest to lowest total error cost
print("Scenarios ranked by total error cost (highest first):")
print(df[['scenario_id', 'accuracy', 'total_error_cost']]
      .sort_values('total_error_cost', ascending=False))

print()  # Blank line between tables.

# show highest to lowest accuracy, to make a comparison
print("Scenarios ranked by accuracy (highest first):")
print(df[['scenario_id', 'accuracy', 'total_error_cost']]
      .sort_values('accuracy', ascending=False))
```

```
Scenarios ranked by total error cost (highest first):
  scenario_id  accuracy  total_error_cost
0          S1    0.9900            500008
3          S4    0.9536            343000
2          S3    0.9930             27140
1          S2    0.9918              7520

Scenarios ranked by accuracy (highest first):
  scenario_id  accuracy  total_error_cost
2          S3    0.9930             27140
1          S2    0.9918              7520
0          S1    0.9900            500008
3          S4    0.9536            343000
```

[3]:
```
fp, fn = 130, 95
cost_fp, cost_fn = 300, 3200

print("S4: Threshold settings ranked by sweep step")
print(f"{'Step':>4} {'False positives':>16} {'False negatives':>16} {'Total␣
 ↪error cost':>18}")

best_step = None
lowest_cost = float('inf')

for step in range(0, 6):
    f_p = fp + 4 * step
    f_n = fn - step
    total = f_p * cost_fp + f_n * cost_fn

    print(f"{step:>4} {f_p:>16} {f_n:>16} ${total:>17,.0f}")
```

```
    if total < lowest_cost:
        lowest_cost = total
        best_step = step

print()
print(f"Lowest cost among tested settings: step {best_step}, "
      f"with total error cost of ${lowest_cost:,.0f}.")
```

S4: Threshold settings ranked by sweep step

Step	False positives	False negatives	Total error cost
0	130	95	$ 343,000
1	134	94	$ 341,000
2	138	93	$ 339,000
3	142	92	$ 337,000
4	146	91	$ 335,000
5	150	90	$ 333,000

Lowest cost among tested settings: step 5, with total error cost of $333,000.